

Inventory and Supply Chain Analytics



Filters

34.08%

Warehouse Utilization

15.56

Days Sales of Inventory

23.47

Inventory Turnover Ratio

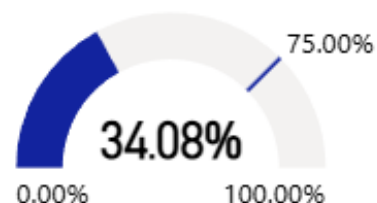
Region

All

Category

All

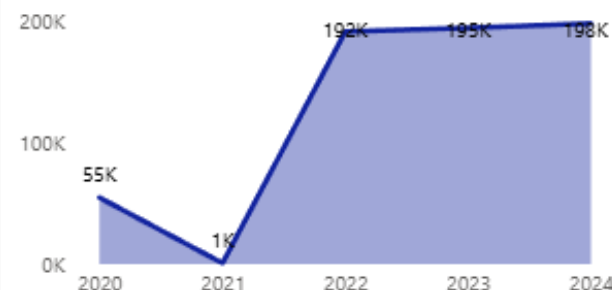
Warehouse Utilization



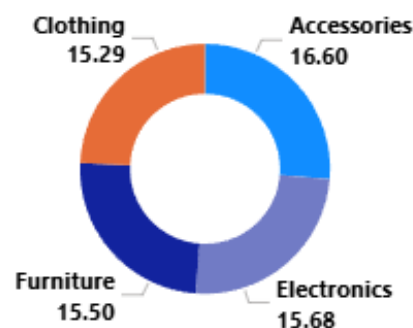
Sum of Transportation Cost by Region



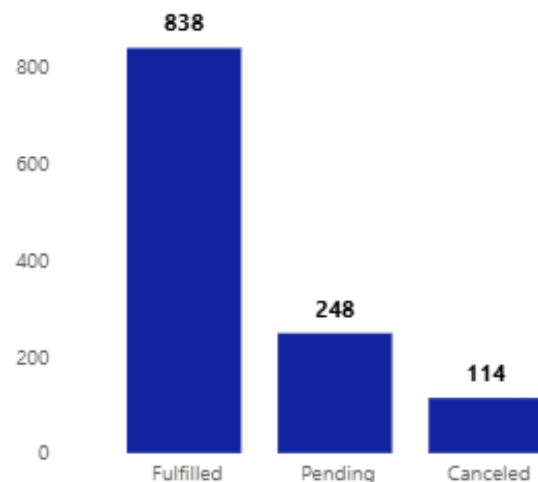
Sum of Units Sold by Year



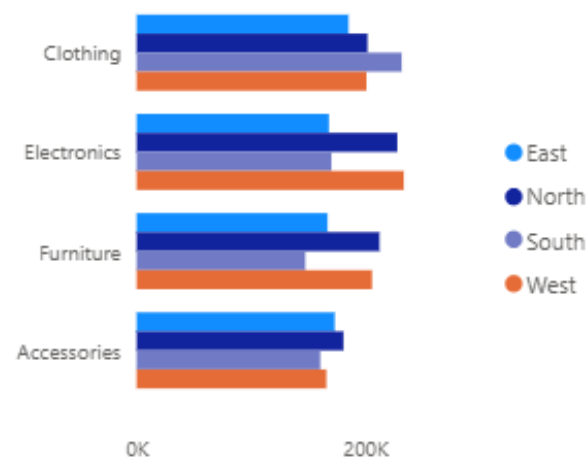
Average of Lead Time by Category



Count of Backorder by Order Status



Sum of Inventory Level by Category



Supply Chain, Page 1

Live data

Data updated on 4/26/26, 3:17 AM



breakdown of what this Inventory and Supply Chain Analytics dashboard tells us

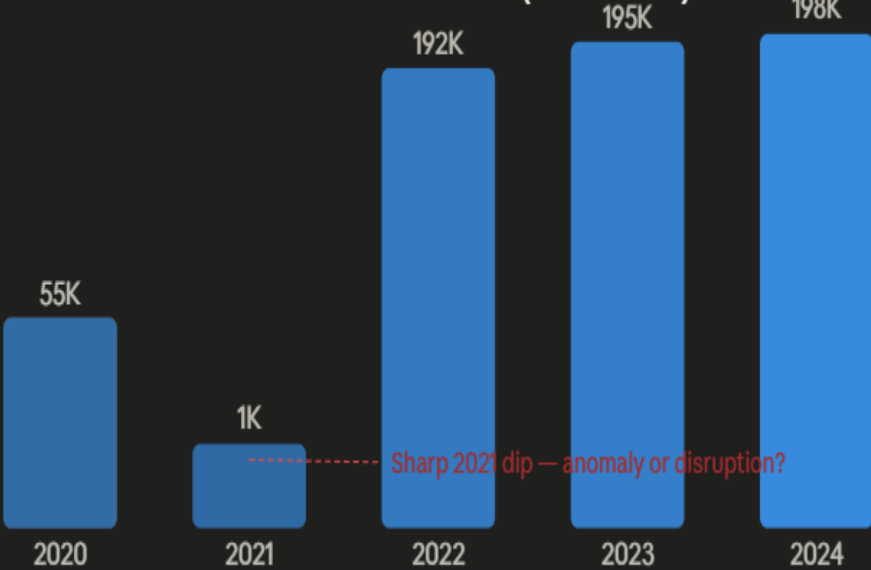
Key performance indicators



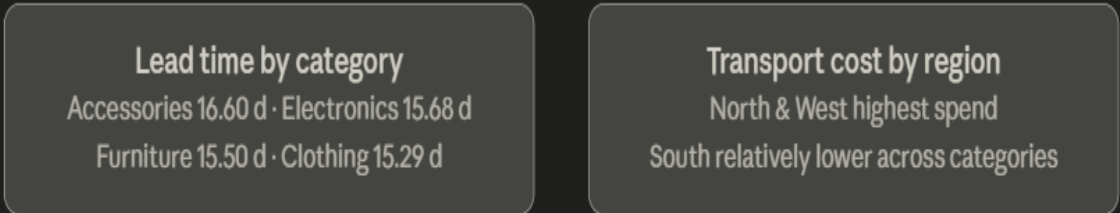
Order fulfilment health



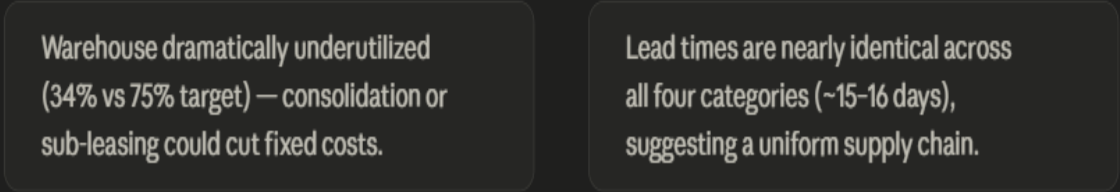
Units sold trend (2020-2024)



Supply chain efficiency



Top takeaways



- **Operational efficiency is strong, but warehouse capacity is a red flag.** The inventory turnover ratio of 23.47× and a DSI of just 15.56 days mean products are moving quickly — that's healthy. But warehouse utilization sitting at only 34% (well below a common 75–80% efficiency benchmark) signals either significant overcapacity, or that inventory is being held elsewhere (3PL, supplier warehouses). This mismatch is worth investigating because underused space still incurs fixed costs.
- **2021 was a serious disruption year.** Units sold collapsed from 55K in 2020 to just 1K in 2021, then rebounded explosively to 192K+ in 2022 and has held steady through 2024. This spike-and-crash pattern is characteristic of supply chain disruptions (potentially COVID-related), a system reset, or a major data quality issue. Any forecasting model must treat 2021 as an outlier.
- **Order fulfillment has a meaningful leakage problem.** While 70% of orders are fulfilled, the 21% pending and ~10% cancelled rate is worth attention. The pending backlog of 248 orders could convert to cancellations if unresolved, worsening that 10% loss figure.
- **Lead times are suspiciously uniform.** All four categories — Accessories (16.6 days), Electronics (15.68), Furniture (15.5), Clothing (15.29) — are clustered within about 1.3 days of each other. This is unusual in real supply chains where bulkier items like furniture typically take much longer. It may indicate standardized supplier contracts, or that lead time data is being averaged rather than measured per-order.
- **North and West regions bear the highest transportation costs.** This likely reflects geographic distance from distribution centers or lower shipment density. The South region's relatively lower costs could make it a target for expanded distribution investment.

Strategic Recommendations

- **Optimize Warehouse Space:** With Warehouse Utilization at 34.08%, there is a significant opportunity to optimize storage costs. We should consider consolidating inventory or exploring "Storage-as-a-Service" models to monetize underutilized space.
- **Improve Lead Time Consistency:** The analysis shows varying lead times across categories (Average ~15.8 days). By investigating the root causes in the Accessories category (highest lead time), we can implement better supplier management to ensure faster fulfillment.
- **Address Backorder Bottlenecks:** With 248 Pending and 114 Canceled orders, focusing on the supply chain "friction points" is crucial. Automating reorder points using the Inventory Turnover Ratio will help reduce cancellations and improve customer loyalty.
- **Regional Cost Control:** The West region shows higher transportation costs across multiple categories. Negotiating better rates with regional carriers or re-evaluating the distribution center locations could lead to significant logistics savings.
- **Capitalize on Sales Trends:** Following the massive recovery and growth since 2021, the focus for 2024 should be on Inventory Buffering during peak periods identified in the "Units Sold by Year" trend to prevent stockouts.